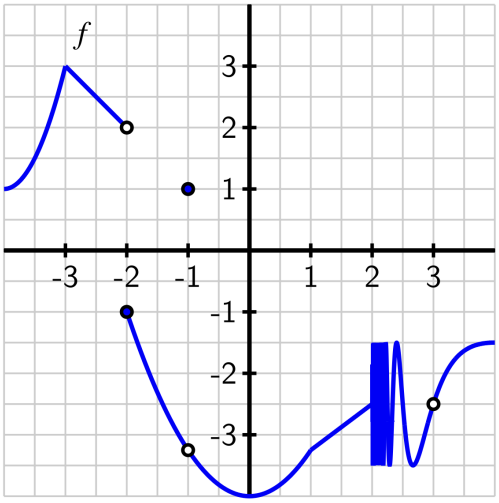
Worksheet 8 - Continuity and Differentiability

MATH 1271

10/01/2026

1. Consider the function $f(x)$ graphed below. You may assume that $f(2) = -2.5$.



   (a) At which values of a does $\lim_{x \to a} f(x)$ not exist?

   (b) At which values of a is $f(a)$ not defined?

   (c) At which values of a does f have a limit (i.e., $\lim_{x \to a} f(x)$ exists), but $\lim_{x \to a} f(x) \neq f(a)$?

   (d) Based on your reasoning above, state all values of a for which f is *not* continuous at $x = a$.

   (e) State all values of a for which f is *not* differentiable at $x = a$. (Give a reason for each of your answers.)

2. (a) Consider the following piecewise-defined function:

$$f(x) = \begin{cases} 3x + 8 & \text{if } x \leq 0 \\ \frac{2}{3}x + \frac{7}{3} & \text{if } 0 < x \leq 1 \\ 4 - x & \text{if } x > 1 \end{cases}$$

i. Compute $f(0)$ and $f(1)$.

ii. For each of the values $a = 0$ and $a = 1$, compute $\lim_{x \to a^-} f(x)$ and $\lim_{x \to a^+} f(x)$.

iii. Based on your work above, is f continuous at $a = 0$? How about at $a = 1$? (Why?)

(b) Consider the following piecewise-defined function, where a and b are constants:

$$g(x) = \begin{cases} a + bx & \text{if } x < 2 \\ 5 & \text{if } x = 2 \\ x^2 - bx + a & \text{if } x > 2 \end{cases}$$

i. In terms of a and b, what is $\lim_{x \to 2^-} g(x)$?

ii. In terms of a and b, what is $\lim_{x \to 2^+} g(x)$?

iii. What is $g(2)$?

iv. What values of a and b make $g(x)$ continuous at $x = 2$?

(c) Determine the value of b that makes the function

$$h(x) = \begin{cases} b - x & \text{if } x < 5 \\ -x^2 + 6x - 6 & \text{if } x \geq 5 \end{cases}$$

continuous at $x = 5$. Clearly explain how you got your answer.

3. We'll use the topics we learned this lecture to explicitly determine the structure of the linear approximation $f(x) \approx a + b(x - x_0)$ of a function $f(x)$ around $x = x_0$. We'll show that $a = f(x_0)$, and $b = f'(x_0)$.

   Throughout this exercise we'll assume that the real numbers a, b, c satisfy:

$$\lim_{x \to x_0} \frac{f(x) - (a + b(x - x_0))}{(x - x_0)} = 0 \tag{1}$$

   and that $f(x)$ is differentiable at $x = x_0$.

   We'll first show that $a = f(x_0)$.

   (a) Assume instead that:

$$\lim_{x \to x_0} (f(x) - (a + b(x - x_0)) = 0 \tag{2}$$

   and use that, along with the continuity of $f(x)$ at $x = x_0$, to show that $a = f(x_0)$.

   (b) How can we use our assumed limit equation Equation 1 to imply Equation 2?

We'll now show that $b = f'(x_0)$.

(a) Now use Equation 1 and that $f(x)$ is differentiable at $x = x_0$ to manipulate the limit and determine that $b = f'(x_0)$. (The limit definition of the derivative will be useful here.)

We could do a similar derivation of the *quadratic* approximation, but that unfortunately relies on a stronger version of the Mean Value Theorem. I'd be happy to discuss that in office hours, however!